

Final Project Request for Proposal

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There is a growing need for mental health services to meet the demand for the rising mental health issues in young adults. Mental health issues are on the rise, yet utilization of mental health resources remains low. This report is aimed at identifying mental health trends and patterns, give data- driven insights and propose a potential solution to the widening mental health treatment gap.

Database Search Process

The first step was to find data that tells the story of rising mental health issues and low mental health resource utilization. The Centers for Disease Control and Prevention website has a Mental Health Channel with databases addressing various mental health related issues across several demographics throughout the United States, so this was the main resource as the agency has several branches that gather different data.

The first dataset is from the National Health Interview Survey (NHIS) branch of the CDC. The dataset contains Mental Health Indicator data to determine utilization of mental health resources across different demographics like age, sex and race/ ethnicity. The dataset has 9 columns and 556 rows. The data contained in applicable from 2019 to present date and was last updated August 27, 2025. The comma separated file of the dataset was downloaded from the NHIS branch of the CDC.

The second dataset is from another branch of the CDC called National Syndromic Surveillance Program (NSSP). This dataset contains data showing Mental Health related emergency room visit rates. Emergency room visits are critical in identifying high traffic health cases that can develop into threats if not addressed. The visit rates are calculated per 100 000 emergency room visits. The mental health related emergency room visits for each demographic

is multiplied by 100 000 ER visits to get the visit rate per 100 000 visits. The data is collected daily from more than 6900 hospitals and is assessed. The last update to the website was November 25, 2025, and this information is applicable from January 2019 to date. The dataset contains 5 columns and 9758 columns. The dataset was downloaded as a comma separated file. See Table 1 and Table 2 in the appendices section for the two data dictionaries.

Data Preparation

The second step is cleaning the dataset for the visualization process. Tableau Prep Builder version 2025.2 was used to clean up the data. For the NHIS dataset showing mental health indicators, I connected the text file into Tableau Prep and added a cleaning step to start the cleaning process. See Figure 3 in Appendix A for the screenshot shows the connected dataset with a clean-up step added. The year had some null values linked with the question “How often do you get the social and emotional support you need?” and race/ ethnicity. The data seems to be missing completely at random as it looks as though the date was not included for the respondents to the survey in sex and race/ ethnicity demographic groups for the US mainland, but it was included for the Pacific Islanders, Native Hawaiians and age- based participants. The decision was made to use lift wise deletion on these rows that are 2% of the dataset and a date is needed to determine how applicable the data is since health information is time sensitive.

The data types for the year column was also changed to year so that Tableau would recognize the column as a year not a number. Data types for percent, Lower CI and High CI were also changed from string to number (decimal) as the columns contained numbers. The percent column, lower CI and high CI also contained null values linked with Pacific Island and Native Hawaiian respondents. Below is a screenshot showing the null values in the percent, lower CI and high CI column.

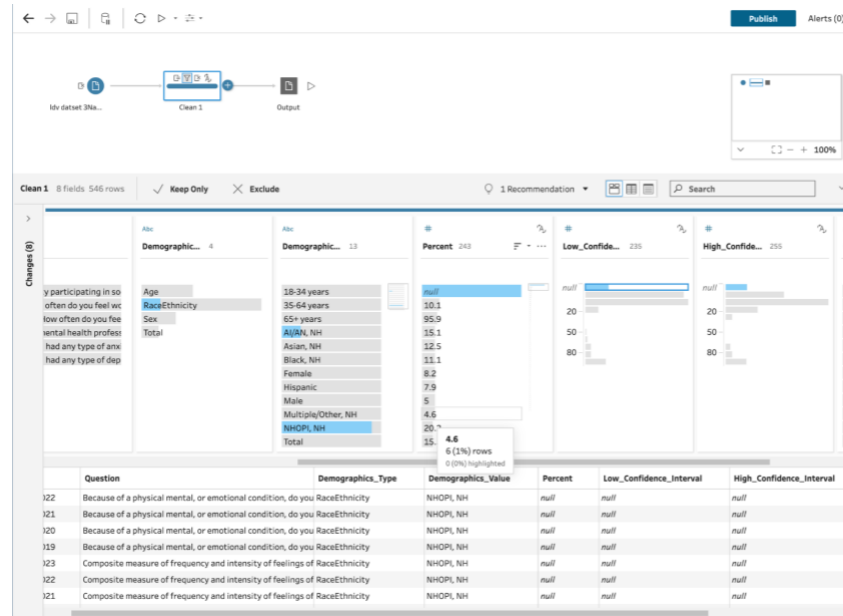


Figure 4: Screenshot to show the null values in the percent, lower CI and high CI columns in Tableau Prep Builder. Taken Dec 1, 2025

The data is missing completely at random as these values were not calculated in the survey. These rows were also deleted as the project's focus is the age- based demographic group not race/ ethnicity. The dataset was also checked for duplicate values, and none were found. Data cleanup is complete and the file was saved as a hyper file format that Tableau Desktop can recognize. See Figure 5 in appendix A screenshot of the hyper file.

For the NSSP dataset showing mental health related emergency room visit rates, I connected the data to Tableau Prep Builder and added a cleanup step. Once the dataset was uploaded onto Tableau Prep Builder I checked for missing and null values. The dataset did not contain any. I also checked for duplicate values and each row was unique. The only change needed was to change the month end column from string to date. The file was then saved as a hyper file. See Figure 6 in the appendix section for a screenshot of the hyper file.

Data visualization

Tableau Desktop version 2025.2 is the visualization tool of choice. The datasets were connected into Tableau and a relationship to connect the two extracts was made to aid in making the dashboard interactive by linking columns. Figure 8 in appendix A shows a screenshot of the creation of an equal relationship between the month end in the ER visits dataset and the year in the utilization dataset. Charts were created by clicking on a new sheet and dragging variables into the pane. The goal was to create charts to show mental health ER visit years over the years, resource utilization over the years and prevalence of each mental health condition.

To do this I dragged month end (dimension) to columns and rate per 100 000 visits (measure) to rows. The numbers on the y-axis looked inflated as it was a sum of the rate per 100 000 visits. To correct this, the sum was changed to average to reflect the true rate. The month end was also changed to month so better depict the trend across the months over the years. Demographics values were dragged to filters and the age-based demographics only were selected as this was the focus group for this project. Demographic values were also dragged to the color mark card so that each demographic age was a different color.

Knafllic advises using one or two colors consistently to reduce the cognitive load on the audience (Knafllic, 2025). To follow best practices, the color of the lines was changed to blue as the highlighting color for the young adults demographic from 18- 34 years as this age group consists of college aged respondents. Black was used to highlight high school aged students 12- 17 years as they had high levels of ER visits that I wanted to highlight and discuss as they had high ER visit rates. The rest of the demographics were grayed out as they are not the highlight of this graph, but they can still be used to help derive insights. Blue and black were chosen as they do not add too much noise to the chart and are good color choices for colorblind audiences.

Demographic values were also dragged to details so that when a point on the line is selected, the rate per 100 000 visits for that demographic shows up. This reduces the cognitive load on the audience as they do not have to try to figure out what each value is by tracking back to the axes. To make the values in the axes more visible I increased the font on the worksheet to 15 and made the text black from gray for easy readability especially when projected on a projector.. I also removed gridlines to avoid unnecessary cognitive load. The x- axis label was taken off as it was adding unnecessary cognitive load and a title was included for the worksheet to label the chart which can help explain what the chart shows at first glance. See Figure 8 in the appendix for the resulting graph.

The second visualization created is a bar chart to illustrate the number of mental health cases reported for each condition per 100 000 ER visits. To create this bar chart, condition (dimension) was dragged to rows and rate per 100 000 visits (measure) was dragged to column. Just like the first visualization, the sum of rate per 100 000 ER visits was changed to average to avoid overinflating values. The names of the conditions are on the y- axis for easy readability by the audience by making the names horizontal instead of vertical. The values were arranged in descending order so that the most common mental health condition is at the top.

Next demographic value was added to filters to filter out all other demographic types except age. Condition was added to color so that each bar is its own color. The colors of the bar were then changed to leverage color as a highlighter by highlighting any mental health issues and anxiety in blue as these were the top two conditions reported for respondents aged 18- 34years the focus group. All the other mental health issues were grayed out. The font for the worksheet was also increased to 15 and text changed to black for easy readability. See Figure 9 in the appendix for the final chart.

The third chart created is a line chart showing the average percentage of mental health resource use each year from 2019- 2024. Percent (measure) was dragged to rows and the year (dimension) to columns. A line chart was automatically generated but because the default Tableau setting used the sum of the percent the numbers were inflated. To fix this I switched the percent which is a point estimate of the percentage to average to make the values more realistic. The year was broken down to a month in the date hierarchy. This did not add much detail to the chart, but the chart was more spread out to allow the trend to be more noticeable especially from 2022 to 2024 where there is a slight change in percentage utilization.

Demographics value was also added to filters to filter out all other demographics except age so that mental health utilization can be analyzed by age just like ER visit rates were analyzed by age. Demographic values were dragged to the color card and blue was used to highlight the 18- 34 age group just like with the ER visit rate chart, and the rest were grayed out as they were not the target age group. Demographics was also dragged to the text card so that when a value on the chart is selected, the percentage utilization for that year shows. The font was increased to 15 and the text was changed to black for easy readability. The y axis label was changed to average utilization rate (%) from just percent as the new label is more descriptive and can enhance audience understanding of the chart. I also added a title for the chart to label it for the audience to understand what they chart shows with minimal cognitive effort. See Figure 10 in the appendix for the final chart.

Once all the charts had been created the next step was to create an interactive dashboard. After clicking create a new dashboard I dragged all the charts into the pane one by one. At first, I was going to use containers for the dashboard but once I started dragging charts into the pane, the containers were overcrowding the graphs making it hard to read the values, so each chart was

switched to floating and dragged to a position on the pane. The ER visit rate chart was placed on the left as that is the chart the audience needed to see first to understand the rest of the charts. I placed the utilization chart next to the ER visits chart to create a connection between need (number of ER visits) and utilization (percentage of mental health resources used). The third chart was placed at the bottom as it is a supporting chart for the ER visits to determine the prevalence of common mental health conditions.

After deciding on chart placement, text boxes were added to highlight key statistics like ER visit rates for 18- 34-year-olds versus those of older adults aged 65+ years as young adults had significantly higher mental health related visit rates. Another statistic that needed highlighting is the decrease in utilization rate from 2021 to 2024 despite the ER visit rates increasing over the years. This statistic is important in painting the picture of increasing mental health issues and growing utilization gap. I highlighted the 14% ER visits percentage increase in blue, bold and enlarged text and the 4% decrease in utilization in black, bold enlarged letters. Blue and black were the colors of choice for consistency with the charts to avoid adding too many colors that could drown the highlighted information. A title was added so that the audience understands that the dashboard is intended to show mental health trends.

To make the dashboard interactive, actions were added to the charts. The first action is for the ER visit rate chart and the mental health cases reported chart. A filter was added so that when a user hovers over any point in the ER visit rate graph the number of mental health cases reported (prevalence of condition) for that demographic for that year shows. This is important to show which condition affects a particular age group the most and allows for condition- specific treatment. For example, 18- 34-year-olds have high cases for any mental health illness this shows that several respondents suffer from mental illness, yet they do not have an actual diagnosis, so

steps need to be taken to get more people diagnosed. The mental health utilization chart uses a filter so that when the user clicks on any point on the utilization chart they can see the ER visit rate numbers for the beginning of the year. This is important to show the utilization and ER visit rate mismatch. Figure 11 in the appendix shows a screenshot of the dashboard.

Key Findings

Firstly, that data showed fluctuating mental health related emergency room visits from 2019 to October 2025. The highest number of visits was from the beginning of 2020 to around February 2021. This rapid increase in number of mental health cases across all demographics is to be expected due to the COVID- 19 pandemic. Thereafter for all demographics the rate of ER visits continued to fluctuate. In 2025 for the age group 18- 34 years there was a slight increase in mental health related visits compared to the previous year end. For this age group, mental health cases had reached their lowest level for the year 2024 but in January of 2025 the visit rate increased by a 100 and has remained slightly high throughout 2025. This shows that mental health related visits are on the rise for young adults. It is also important to note that high school aged students (12- 17years) consistently have the highest rate of ER visits amongst all demographics.

To add on to that, mental health resources were used the most from 2019 to 2021 which was during the pandemic and post pandemic era. From 2021 going forward the percentage of utilization dropped by 4% for 18- 34-year-olds. The percentage of mental resources used since 2022 has remained around 28%. This shows that while the number of mental health related ER visits fluctuates and increases, utilization remains consistently low creating a gap between utilization and mental health resource need.

Another important finding is that for the young adults (18- 34 years) the highest cases reported were for any mental health issues and anxiety. Most people surveyed reported having mental issues but could not name the mental issue suggesting that several people are living with mental illness without a specific diagnosis and are not getting condition- specific treatment. Anxiety was the second most common case reported in young adults (18- 34 years).

Proposal

A potential fix for the rising mental health issues and low resource utilization alongside with the high numbers of people living with unspecified mental issues and anxiety is a digital self- screening tool. Calm health is a digital app that is subscription based and it offers personalized mental health plans for users. The app is invite only by a counsellor, or health professional which allows for patient privacy (Calm Health, n.d.). Users complete a mental health screen by answering a set of questions and set mental health goals. Based on these results the user is made aware of mental health issues they are dealing with, and they get recommendations for mindfulness techniques like exercises, sleep stories and counselling through completion of programs designed by psychologists.

The proposal is for universities like the University of North Texas to pay for the Calm Health subscription for students and give students free access to the app with enrollment. The app would be linked under the counselling services page and resources on the school website for all students to see. At new student orientation, this resource can be shared with students so that they are aware that the app exists. Sharing the app at new student orientation is crucial to equip incoming freshmen with mental health resources as high school students have been identified as having the highest rate of ER visits. The student counselling center can also recommend the app to any patients that they see at their office.

Potential impacts

Introducing students to calm health can allow for early screening of mental health related issues. The mental health screening tool in Calm Health can reveal mental health issues like stress, and anxiety which a college student may not be aware of allowing for early intervention before a mental health crisis. In addition, more students can get a diagnosis and get the care they need, making the University a safer and healthier campus. Raising mental health awareness by offering mental health resources like Calm Health can help reduce the stigma around mental illness by allowing for open and honest conversation relating to mental health around campus.

Conclusion

This project utilized two major datasets from the CDC website to show the widening mismatch caused by rising mental health issues in young adults 18- 34 years and consistently low mental health resource utilization. By showing these trends, key findings, and proposing a data- driven digital self- screening tool the hope is for this proposal to guide in advocating for additional mental health resources for college students to get the care they need before a crisis occurs.

References

App.calmhealth.com. (n.d.). <https://app.calmhealth.com/>

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https://data.cdc.gov/Mental-Health/National-Health-Interview-Survey-NHIS-Mental-Health/d89q-62iu/about_data

Centers for Disease Control and Prevention. (2025, November 25). *National Syndromic Surveillance Program (NSSP) mental health-related emergency department visit rates.*

Centers for Disease Control and Prevention. https://data.cdc.gov/Mental-Health/National-Syndromic-Surveillance-Program-NSSP-Menta/eze9-ahe5/about_data

Appendices

Appendix A: Data Dictionary & Screenshots

Table 1: NHIS Mental Health Indicators data dictionary

Column name	Description
year	survey year
question	question asked
demographic type	respondant demographic (age, Gender)
demographic value	Demographic category- age ranges of people surveyed
	12- 17 years
	18- 34 years
	34- 64 years
	65+years
percent	point estimate for health care usage
low_confidence_interval	Low 95% Confidence interval
high_confidence_interval	High 95% confidence interval
confidence_interval_formatted	Confidence Interval
data source	source of data

Table 2: NSSP Mental Health related emergency room visit rates data dictionary

Column name	Description
month_end	month of year
condtion	mental health condition
demographics_type	demographic type of people surveyed (sex, age, race?ethnicity)
demographics_values	Demographic category- age ranges of people surveyed
	12- 17 years
	18- 34 years
	34- 64 years
	65+years
rate_per_100000_visits	Number of mental health visits per 100 000 hospital visits

Figure 1: Screenshot to illustrate downloading dataset from NHIS. Taken Dec 1, 2025

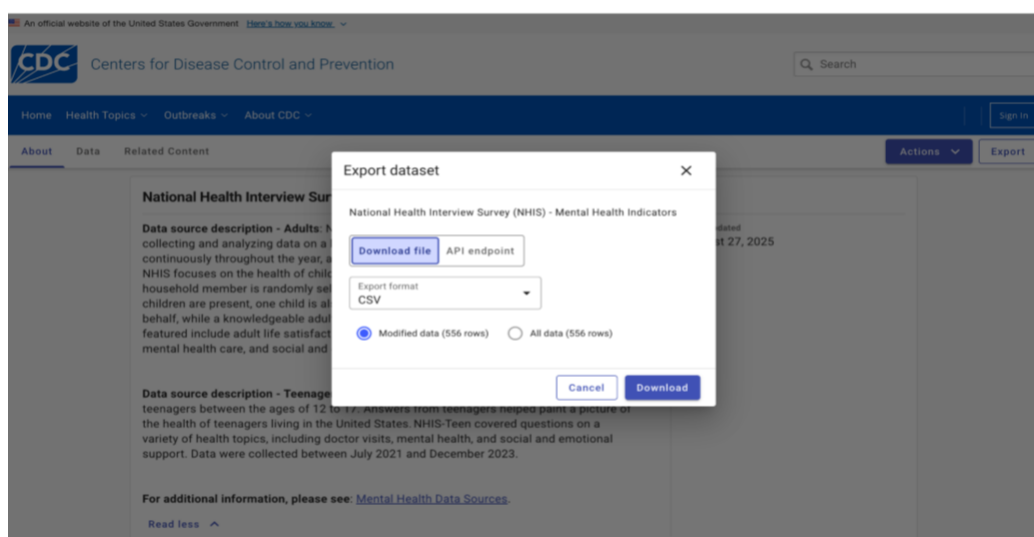


Figure 2: Screenshot to illustrate downloading dataset from NSSP. Taken Dec 1, 2025

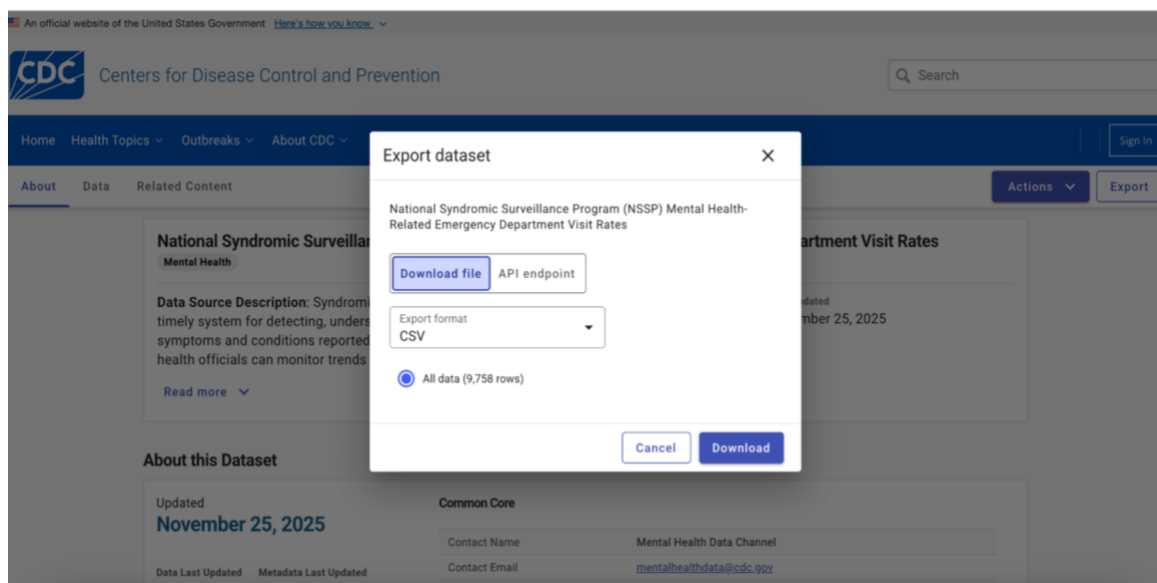


Figure 3: Screenshot to illustrate connecting dataset to Tableau Prep Builder and creating a clean-up step. Taken Dec 1, 2025

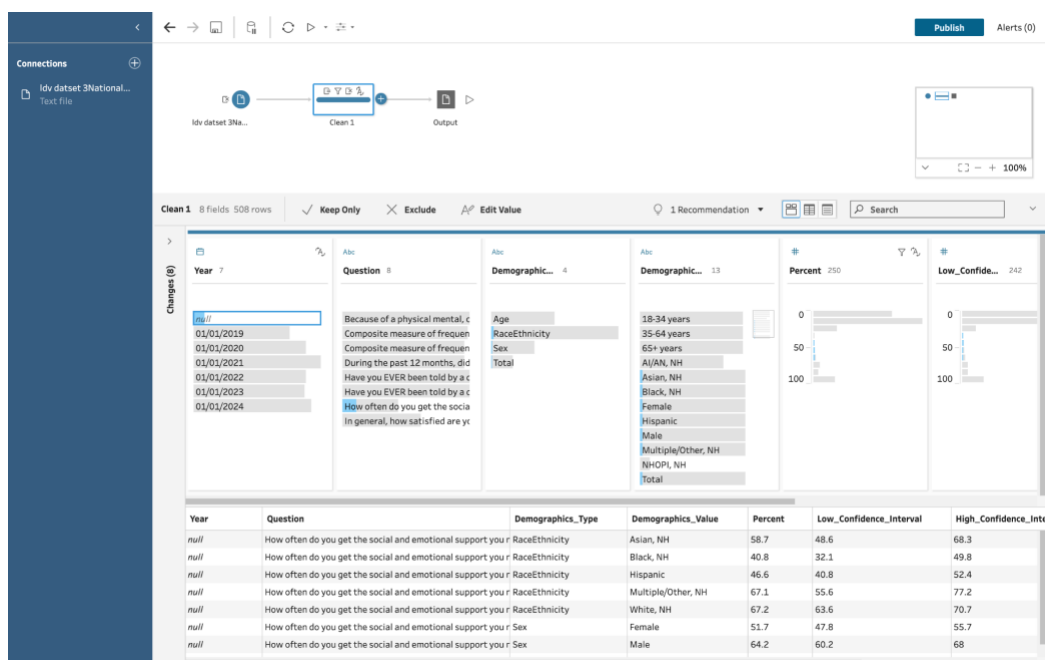


Figure 5: Screenshot to show the hyper file for the NHIS dataset in Tableau Prep Builder. Taken Dec 1, 2025

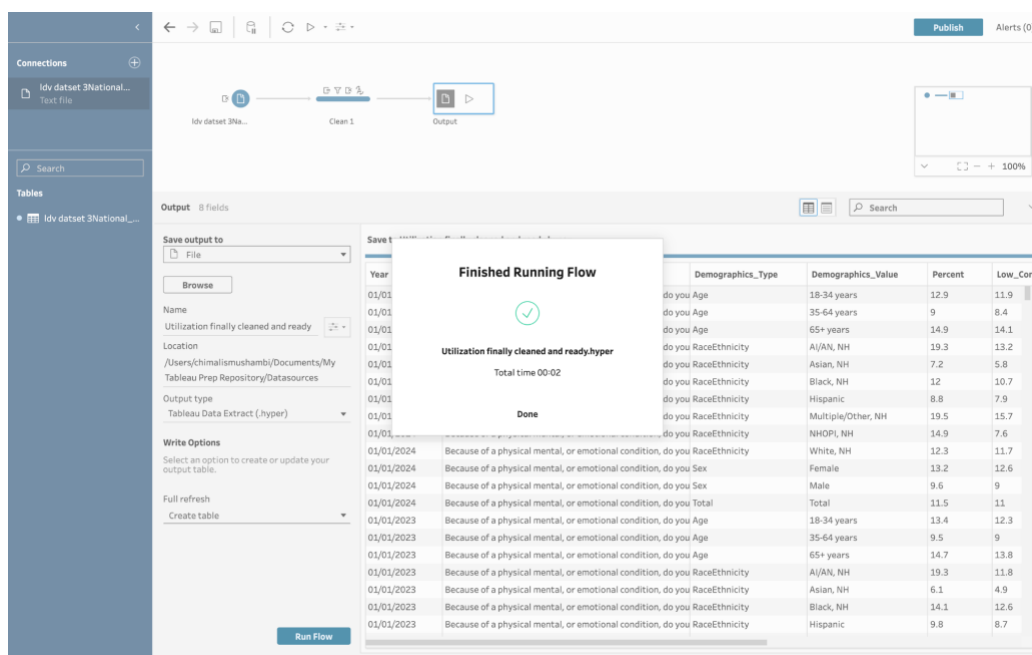


Figure 6: Screenshot of cleaned hyper file for the NSSP dataset in Tableau Prep Builder. Taken November 30, 2025.

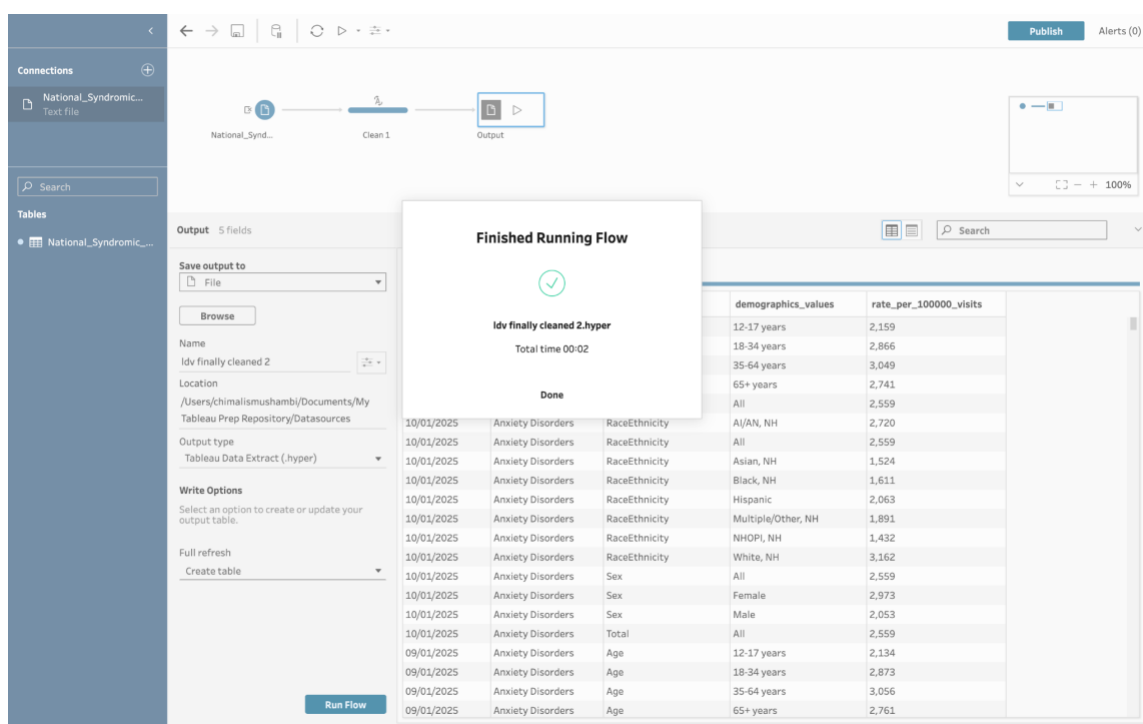


Figure 7: Screenshot showing connecting the two extracts in Tableau Desktop 2025.2. Taken December 1, 2025.

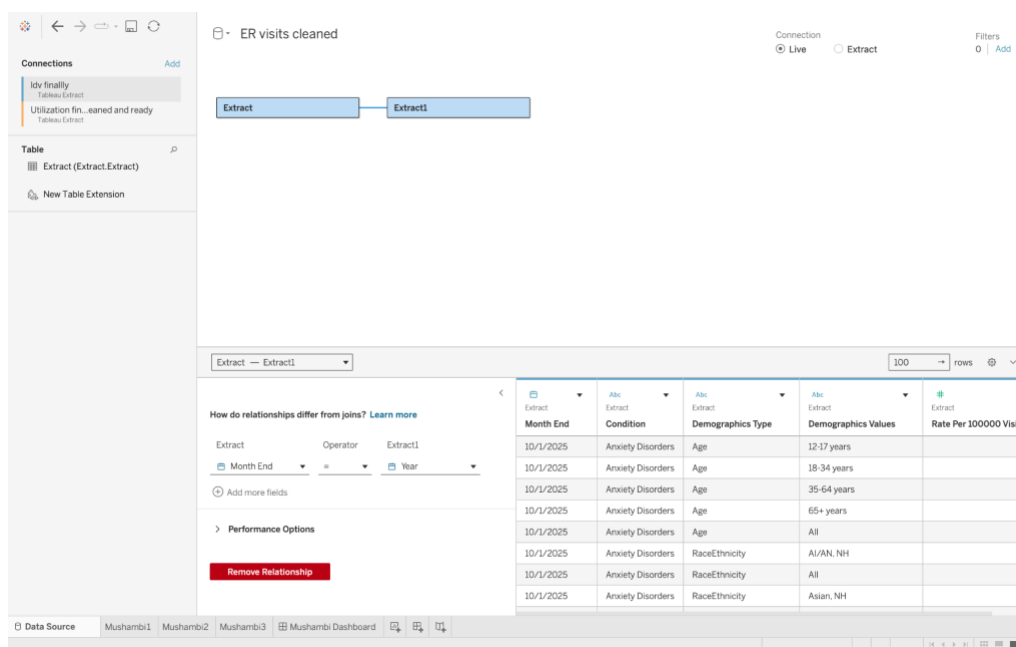


Figure 8: Screenshot showing the final line graph showing avg. mental health related ER visits per 100 000 ER visits in Tableau Desktop. Taken November 28, 2025.

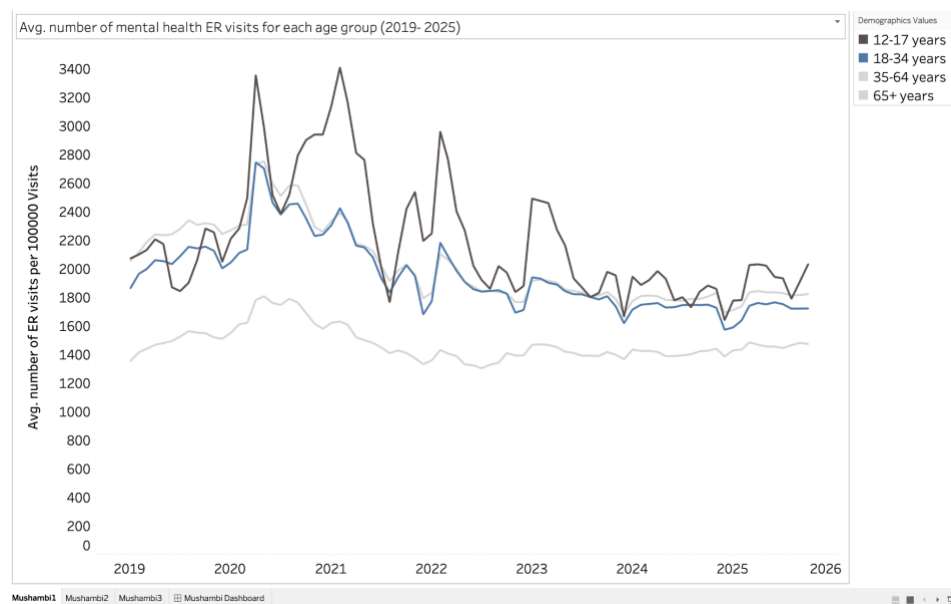


Figure 9: Screenshot showing a bar chart of the avg. number of mental health cases reported for each condition per 100 000 visits in Tableau Desktop. Taken November 28, 2025

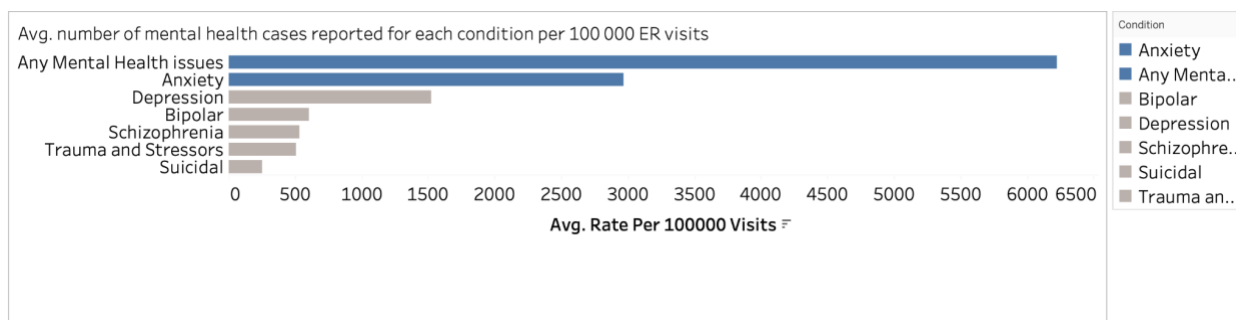


Figure 10: Screenshot showing avg. percentage of mental health resource use per year (2019- 2024)

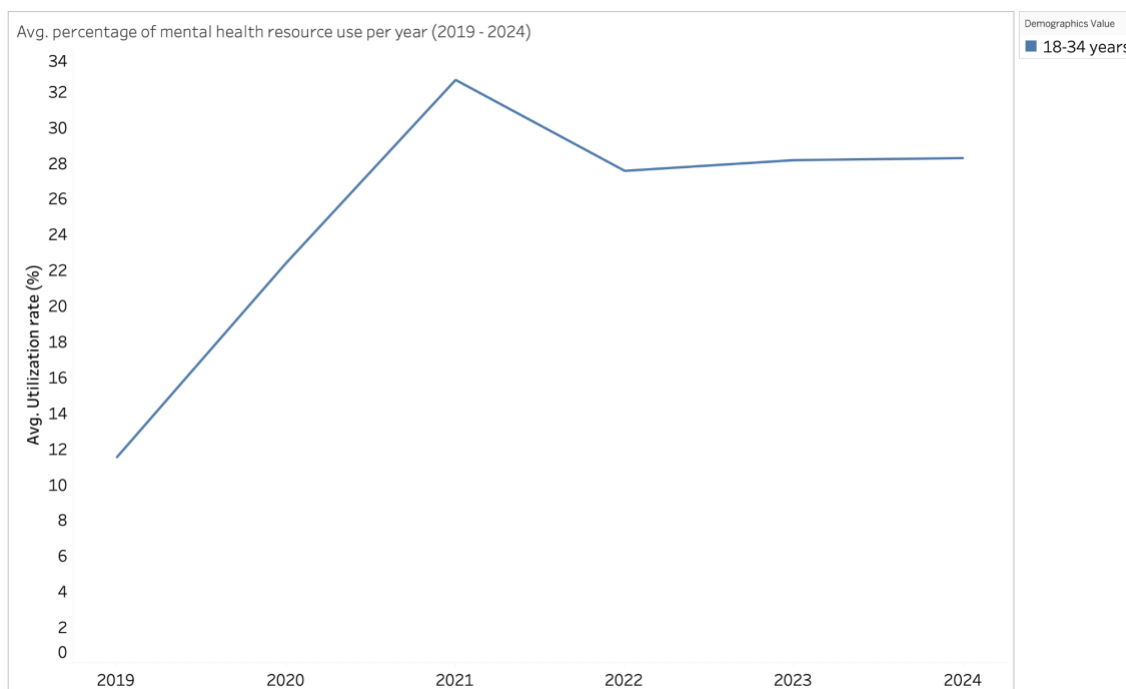
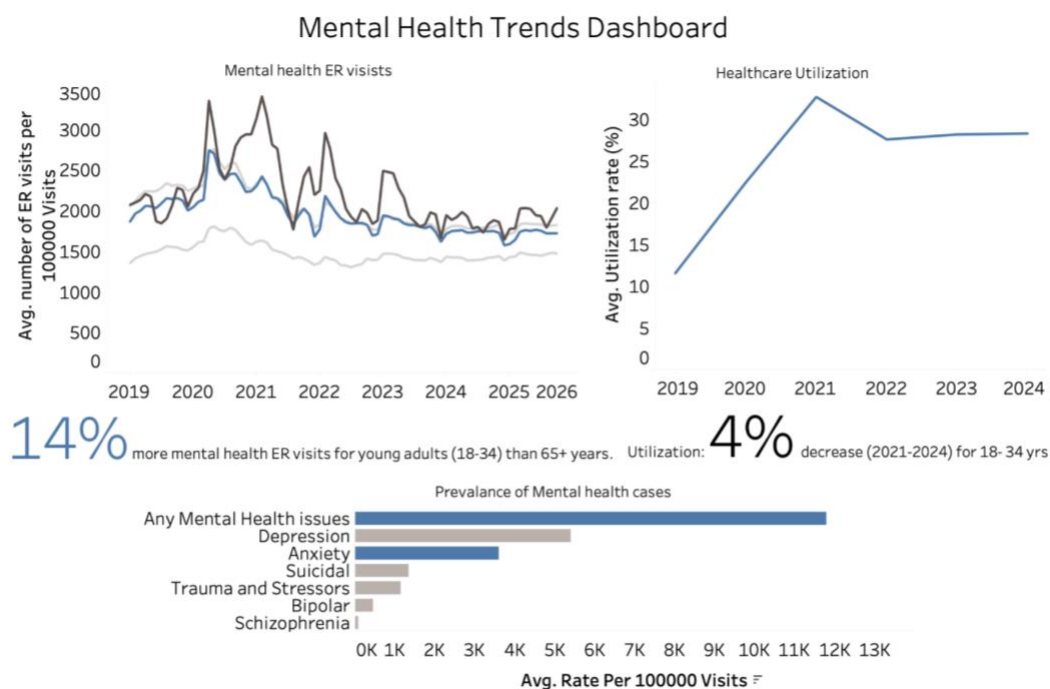


Figure 11: Screenshot of interactive Mental Health Trends Dashboard in Tableau Desktop. Taken Dec 01, 2025



Appendix B: AI Appendix

1. **Why-** to find a research topic

Prompt- major issues that affect college students

AI response- Google search resulted in several articles

<https://www.collegenp.com/article/most-common-problems-faced-by-students>

<https://health.clevelandclinic.org/mental-health-in-college-students>

<https://bau.edu/blog/problems-college-students-face/>

Critique- I liked all the research articles as they pointed out different issues that affect college students like mental health issues and financial issues. These two were common issues from all the articles I read so I decided to go with mental health as a crisis as my research area because Cleveland Clinic is one of the biggest research hospitals in the nation so I could trust the information that mental health is indeed an issue affecting college students.

2. **Why-** Find datasets for use in the project.

Prompt- where can I find databases that talk about stress. real life databases.

AI response- Great — here are several real-life open datasets that cover stress, mental health, and related behaviours. You can explore and filter them to find ones that suit your “hidden stress/burnout” topic.

 Recommended databases

a) COVIDiSTRESS Global Survey

Description: Very large global survey ($\approx 173,426$ participants across ~ 179 countries) that measured perceived stress (PSS-10), loneliness, behavioural responses to the COVID-19 pandemic.

Why it's useful: Key stress variables + many demographics + cross-cultural data.

Use case: Filter to young adult respondents, examine stress levels, compare to other factors (social support, isolation).

Limitations: Pandemic context may influence stress in specific ways.

- b) Databases from Centers for Disease Control and Prevention (CDC) & other U.S. government sources

Example: The CDC's "Mental Health Data Channel" lists surveys, data sources on stress, mental health, etc.

Also: Their children's mental health data section includes prevalence of anxiety, stress indicators among youth.

Why it's useful: Nationally representative, good for benchmarking and broader context.

Use case: Use state- or age-specific data to compare young adult stress/burnout across demographics.

Limitations: May not always have detailed "burnout" variables, might be more general stress/mental health.

- c) Substance Abuse and Mental Health Services Administration (SAMHSA) Public-Data

Description: Offers many datasets on mental health, substance use, related behaviours.

SAMHSA

Why it's useful: Mental health often overlaps with stress/burnout; you might find data on "emotional distress", "work-related stress".

Use case: Use as auxiliary dataset when looking at burnout in young adults, especially related to behaviour or comorbidities.

Limitations: May need to filter carefully to get stress and young adult subsets.

d) Dataset directories & repositories

For example, a paper lists major open datasets relevant to youth mental health and stress.

Also: Platforms like Kaggle host many mental health / stress datasets.

Kaggle

Why it's useful: Great for finding more niche datasets (e.g., young adult specific, academic environment, burnout).

Use case: Search for "stress", "burnout", "young adults survey" etc on these platforms.

Limitations: Varying quality, may require more cleaning/pre-processing.

Critique: I liked the suggestion to look into CDC and SAMHSA as they are US based, government funded websites which I found to be trustworthy. I dismissed the Kaggle suggestion as I asked for real life databases. Subsequent browsing on the CDC and SAMHSA website gave me my datasets for the project.

3. **Why-** To help find a captivating title for my presentation.

Prompt- good title for presentation for mental health awareness and advocating for digital self-screening test on campus.

AI response- Here are polished, strong title ideas for a mental health awareness presentation that also advocates for digital self-screening on campus:

"Your Mind Matters: Bringing Digital Mental Health Tools to Campus"

"Pause, Reflect, Screen: A New Approach to Campus Mental Wellness"

"From Silence to Support: Why Students Need Digital Self-Screening"

"Bridging the Gap: Digital Screening as a Path to Early Mental Health Support"

"See the Signs Sooner: Advocating for Digital Self-Screening on Campus"

"A Smarter Way to Support Students: The Power of Digital Mental Health Checks"

Critique- I did not find any topic that I liked but the topics helped me find words to start drafting potential topics for my research proposal presentation.

4. Why- to find a digital mental health screening tool

Prompt- example of mental health digital screen tool used by a college

AI response- An example of a digital mental health screening tool used by colleges is the CCAPS-Screen, a tool that measures stress in areas like depression, anxiety, and academic distress, providing personalized resource recommendations. Other examples include apps like Calm Health, which uses the PHQ-9 and GAD-7 screening tools to suggest resources based on student responses, and platforms like WellTrack Boost.

Critique- I like the suggestion to look into the Calm Health questionnaire as the proposed digital tool for my proposal as it uses board certified mental health screening tools backed by the American Mental Health Counselors Association. I like how it is anonymous since it is by invitation only making it perfect for my proposal as anonymity is important.